

Relay Attacks in Intra-handshake Attestation

Muhammad Usama Sardar^{1,2}, Viacheslav Dubeyko³,
and Jean-Marie Jacquet⁴

(ACK: Juho Forsén, Mariam Moustafa, Markus Rudy, Tjaden Hess,
Yuning Jiang, and Pavel Nikonorov)

¹TU Dresden, Germany

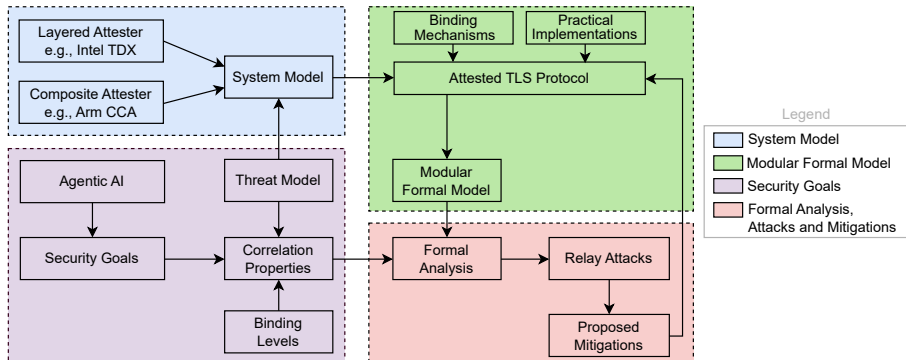
²Co-chair, Trusted Research Environment (TRE) Open Suite,
Global Alliance for Genomics and Health (GA4GH)

³IBM, San Jose, CA, USA

⁴Nadi Research Institute, University of Namur, Belgium

January 13, 2026

Approach

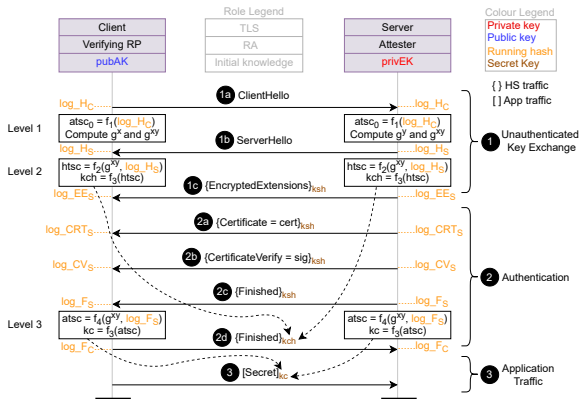


Potential mechanisms for binding (server as Attester)

Binding material	Value of rdata field in formalization
Client's attestation nonce	na
Client's TLS nonce	nc
Early exporter	exp_0
(Hash of) Server's public key	pubEK or hash(pubEK)

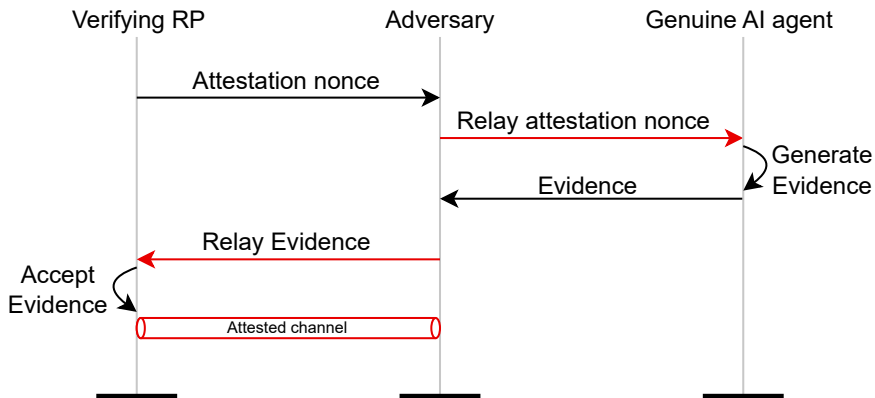
- Combinations
 - Hash (Client's TLS nonce || Server's public key)
 - Hash (Client's Attestation nonce || Server's public key)
- **Discussion:** Any other binding material/combination?

Strong Binding vs. Relay of Evidence (Abstracted)

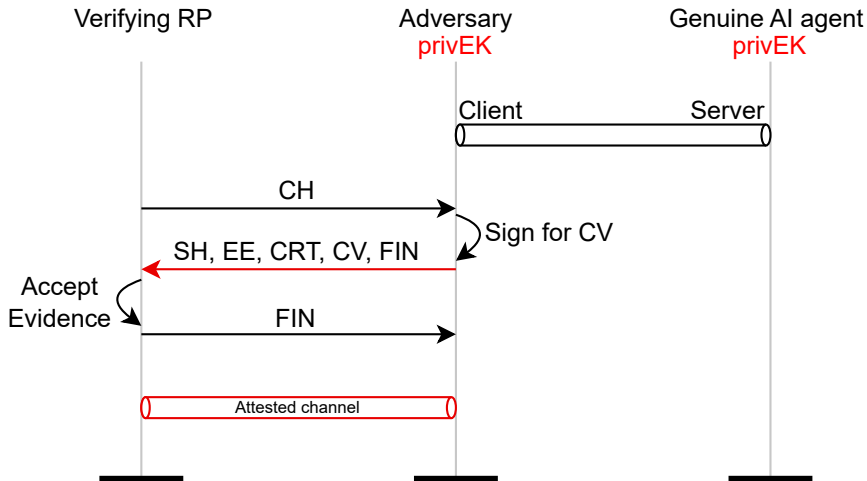


- $htsc$: used for encryption of clientFinished message (2d).
 - Irrelevant for security goals
- $atsc$: used for encryption of application data (client's secret, e.g., decryption key)
 - Relevant for security goals

Relay Attack 1 (Abstracted)



Relay Attack 2 (Abstracted)



Implementation Issues

- Meta's AI: TLS nonce (hence, no Evidence freshness)
- Conveyance of attestation nonce
 - Edgeless Systems Contrast (previously) and Cocos AI abuse SNI (intended for hostname)
 - Edgeless Systems Contrast (now) abuse ALPN (intended for protocol names)
- CCC Attested TLS proof-of-concept: No update for 3 years
- **Discussion:** Any other intra-handshake attestation implementation?